rates and intra-story trigram repetition rates (Yao et al., 2019). The former is a diversity metric - if stories look fine but inter-story repetition is high, it means the language model has learned to tell only very similar stories even when conditioned on diverse prompts. Intra-story trigram repetition is a fluency metric, and measures the proportion of trigrams within a single story that are repeated.

For all of these metrics, there is a tension between diversity metrics, which bring a generation closer to human quality, and fluency metrics, which can degrade as diversity increases.

Human Metrics. We run two separate experiments to measure improvement in our target areas of Relevance and Overall Quality. Relevance is defined as whether stories both relate to the given prompt and remain on topic for the duration of the story. Overall Quality is defined to be combination of *coherence*, *interestingness*, and *relevance*, similarly to most other story generation work.[14] We postprocess stories for human review by detokenizing[15], removing special end-of-sentence tokens, and truncating to 250 (whitespace separated) words. We have Mechanical Turk workers rate all systems' outputs on the same prompt comparatively on a *likert score* (1-5) across both metrics. But since likert scores are well known to exhibit a central tendency bias, it is likely to be unreliable on distinguishing between systems that are close in performance, particularly as reading 5 long generations introduces significant cognitive load. Therefore, we further conduct *pairwise comparisons* between the top 3 systems in the likert experiment.

Test Data Selection. In contrast to previous work on this dataset, our 110 human titles are randomly sampled from a *filtered version* of the test set. Writing Prompts has a one-to-many relationship between the prompts and stories.[16] The dataset also contains an artifact of the sort of topic that is upvoted on reddit[17], so many test prompts are minor variations on the same topic. We hypothesise that some of the gap between reported performance in papers on this dataset and performance in the wild is due to the artificially high similarity between training and test prompts, so we randomly sample from the test set, but exclude prompts with extremely high lexical overlap with training prompts.[18]

[14]Surveys used for both may be found in Appendix E, as well in in the code repository.

[15]We use MosesDetokenizer (Koehn et al., 200)

[16]Of the 303358 prompts, only 1/3 (107665) are unique

[17]Mostly aliens

System	Voc:Tok ratio	Entities	Avg Tok
Naive Plot	1.52	8.25	199
Aristotelian Plot	1.81	7.49	168
Gold Plot	3.59	9.26	371

Table 4: Metrics for plots

System	Automatic Evaluation			
	Voc: Tok↑	Diverse Verb %↑	Intra-Rep↓	Inter-Rep↓
Mao et al. (2019)	6.6	81.8	5.68	27.5
Guan et al. (2020)	2.3	71.9	0.60	56.1
Prompt-to-Story	1.5	68.9	0.22	65.1
Naive Plot	1.4	76.4	0.11	63.8
Aristotelian Plot	1.5	74.8	0.12	64.1

Table 5: Automated metrics for all models.

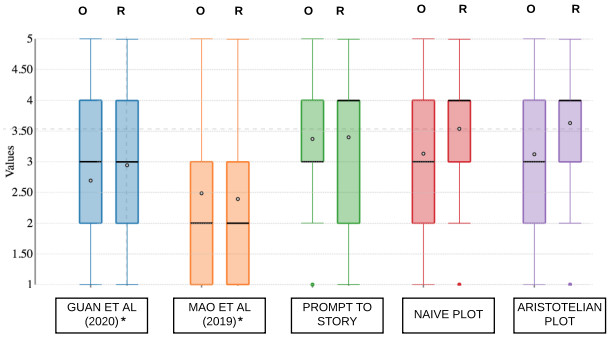
5 Results

Plots. As is shown in Table 4, the Aristotelian plot brings the generated plot structure closer to the gold plot Vocab:Token ratio, though there is still a large gap. This improvement comes at a slight expense of number of entities per plot, which is likely because rescored plots are on average shorter.

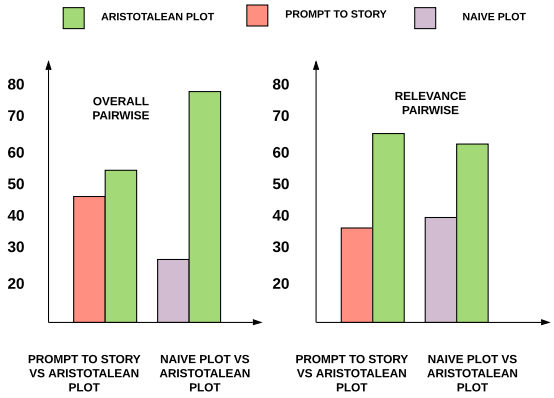
Stories. The combination of automatic and human evaluation for stories highlights both the coarseness of automatic evaluation metrics and the blind spots of human evaluation (Hashimoto et al., 2019). From just the lexical metrics, the two baseline systems of Mao et al. (2019) and Guan et al. (2020) would appear to have the best performance (though Guan et al. (2020) suffers in Diverse Verbs), and the three BART-based systems are indistinguishable save an improvement in Diverse Verbs with the introduction of plot structures, as observed in the plot experiments of Fan et al. (2019).[19] However, intra-story trigram repetition reveals the weakness of the Prompt to Story system and the Guan et al. (2020) system. It also shows the extreme lack of fluency in the Mao et al. (2019) system, which exhibits an unnatural level of repetition characteristic of degenerate output.

[18]72% of prompts were excluded. We used sequence-matcher `https://docs.python.org/3/library/difflib.html` and spacy vector similarity `https://spacy.io/` to exclude prompts with a similarity of 1 to any prompt in the training data when stopwords are removed. The two systems gave identical results.

[19]Though we do observe a similar magnitude increase in Diverse Verbs with the introduction of plots, our baseline Prompt-to-Story model has a higher % than their best model, reflecting the recent performance improvements in pretrained language models. We report Unique Verbs for comparability, but do not find it to be useful metric since it is not normalised by length or token count.



(a) Likert ratings for all 5 systems scored together. O is Overall, R is Relevance, solid bars are median, dots are mean. * denotes systems with significant differences ($\alpha < 0.05$, Wilcoxon signed-rank test). The three BART based systems at right are not statistically significantly distinguishable via this setup due to central tendency bias.



(b) Pairwise win rates for the Aristotelian Plot system and two best baselines. Significant ($\alpha < 0.05$) via Wilcoxon signed-rank test.

Figure 2: Human evaluation for five systems on 110 prompts, with each sample evaluated by 3 judges. The scores for the likert experiment are averaged over all the instances; the pairwise comparison takes the majority vote of the three judges for each instance.

Human Evaluation. The human metrics in Figure 2a then reinforce that the high lexical scores of the Guan et al. (2020) and Mao et al. (2019) systems are a result of those models sometimes deteriorating into nonsensical text, as both of those systems have extremely low human judgement scores.[20] The likert scores favor the Aristotelian Plot system with regard to relevance but favor the Prompt to Story baseline for Overall Quality - though as can be seen from the variance in 2a, all three BART systems are too close together to be reliably distinguished via likert metrics. The pairwise comparisons in Figure 2b do differentiate the three systems with strong statistical significance, showing the superiority of Aristotelian Plot over both the Prompt to Story and Naive Plot systems with respect to both Relevance and the Overall Quality of the final stories.

6 Analysis

We analysed the patterns in reported user confidence and found that 8% of prompts are low user confidence (<3) and 8% are high confidence (>4.5), so we look further into these as examples where the top three systems are minimally and maximally distinguishable. We include examples of outputs for these prompts for all three models in Appendix C.

For Overall Quality, there are no low confidence titles, and the Aristotelian plot system is preferred for all of the high confidence ones. In Relevance, the low confidence prompts have split win-rates across all models (essentially random) and the prompts show that all systems struggle with the meta-level concepts. The Writing Prompts dataset varies from concrete (*A story of a cat who saves a girl from her depression*) to prompts requiring other types of meta-knowledge. Prototypical low confidence titles are of two forms: 1) *A day in this life in a world where everything is written in poetry*, and 2) *Write a story where I can't tell whether the protagonist is a hero or a villain*. The type 1 stylistic instruction prompt requires knowledge about the distinction between instructions for style and content, and all models fail. The type 2 meta-content type of prompt requires a finer level of control of the structure of the plot and story than the Aristotelian system or any of the other models can manage. This kind of prompt presents an interesting case for future story generation work, as it is simple enough for human authors to be popular on a forum, but far beyond the capabilities of current language models. At the high confidence Relevance prompts, the Aristotelian system wins all but one. Those stories highlight the way that adding and then improving on plot structures assists relevance by keeping a story on topic throughout an entire generation (see Appendix C Table 9).

To assess the cases where the Aristotelian Plot did not improve over the BART baselines, we measure both word and verb incorporation rates, in

[20] Examples from each baseline system for the prompt in Table 1 can be found in Appendix D

System	Word Incorp %	Verb Incorp %
Naive Plot	75.5	75.1
Aristotelian Plot	72.4	72.0

Table 6: Metrics for incorporation rates for stories.

Table 6[21]. These measure the Levenshtein distance over the sequence of plot words (excluding verbs) or plot verbs that are included within the final story. While the high incorporation rates show that the story model does utilize the plot, there is a gap between the current utilisation and the possible upper bound. The focus of our work is on improving plot generation, but we hypothesise that modifications to the story model to improve incorporation rates would further widen that performance gap between the three systems, as it will give the plot more influence over the story surface realisation and ensure that plot improvements appear downstream.

7 Related Work

Story Generation without Plots. Diverse efforts have focused on generating stories. Fan et al. (2018) re-purpose an approach for Neural Machine Translation to translate from prompt to a story via Convolutional Seq2Seq models. Guan et al. (2020); Mao et al. (2019) use a similar approach, however they incorporate structured commonsense knowledge from external datasets or knowledge bases to improve a story generated from a prompt. **?** add control to the story ending valence.

Story Generation with Plots. Riedl and Young (2010) use refinement search as a technique to balance between character and plot for solving the narrative generation problem. Li et al. (2013) use plot graphs for story generation that model the intended logical flow of events in the virtual world as a set of precedence constraints between plot events. Martin et al. (2018) decompose the problem of story generation into generation of successive events (event2event) followed by generation of natural language sentences from events (event2sentence). **?** follow up this work by comparing five different models for event2sentence realisation, and find all to have different weaknesses. Yao et al. (2019) improve their LSTM-generated ROCStories by extracting keyword-based plot-like structures, or *storylines*, and using these in a pipeline to generate storylines and then stories. Fan et al. (2019) experiment with numerous techniques for representing story plots on the WritingPrompts dataset, and find Semantic Role Labelling with Entity Anonymization and Compression to work best. More recently, **?** propose a goal-driven reinforcement learning approach to plot generation, and Chen et al. (2019) propose a latent variable model to learn how to generate outlines for neural story generation.

Learning Story Aspects via Rescoring. Holtzman et al. (2018) generate continuation sentences from context sentences, and introduce using a mix of collaborative discriminators that each learn one Grician maxim of conversation, and use them to rerank their RNN story model output. Goldfarb-Tarrant et al. (2019) use those discriminators with the system of Yao et al. (2019) as part of a collaborative story writing task with an LSTM and human writers. However, none of them apply rescorers to the plot. There is no other work that uses discriminators or rescores for plot structures or to try to train them based on different principles.

8 Conclusion and Future Work

We have shown that Content Planning via an interim *plot structure* representation can be combined with the use of rescoring models to inject Aristotelian story-writing principles into the plot. We found that this results in stories that are both more relevant and higher quality than stories that are generated directly from prompts or that use plots without Aristotelian rescoring. Our findings also suggest future work on additional ways to incorporate story principles into plot generation. Although our Aristotelian plots improved over the naive plot, there remains gaps in quality between generated and gold plot structures. There is also further work to be done in investigating what models are best able to incorporate plots, which would enable plot improvements to be even more effective.

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[21] Yao et al. (2019) also use word incorporation rates, but theirs are not comparable to ours, as we both include verbs as their own separate metric (due to their importance in our structure) and make this an *ordered* metric (rather than set intersection as they do), which is necessary because our generated stories are much longer